Jack Goh Pin Kai

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PROFESSIONAL SUMMARY

Computer Science undergraduate with software engineering experience in C++, Python, JavaScript and React.js. Skilled in graphics programming, interactive simulations, full-stack application development, debugging and requirements analysis. Experienced in collaborating with technical teams and translating user needs into reliable software solutions.

KEY SKILLS

- **Programming:** Python, SQL, C++, C#, Java, JavaScript, PHP
- **Software & Web Development:** React.js, TypeScript, Three.js, Vite, Laravel, HTML, JSON, RESTful APIs
- **Data & Analytics:** MySQL, Database Design, Data Modelling, Machine Learning (coursework)
- **Development Tools:** Git, GitHub, Visual Studio, Android Studio
- **Systems & Graphics:** OpenGL, GLSL, ImGui, Entity Component System (ECS), Interactive 3D
- **Tools:** PowerApps, Microsoft Office, Adobe Creative Suite

PROJECTS

VectorSpace | Software Developer | 2026 – Present

- Industry Project - VThere (Client) x DigiPen
- Collaborating with a team to develop a no-code 3D authoring platform for VThere using React, TypeScript, Three.js and Vite.
- Developing camera navigation, object selection, 3D transforms and orthographic views.
- Implementing scene management and editor interactions for interactive 3D environments.

Pulse Protocol | Graphics Programmer | 2025 - 2026

- Engineered modular C++/OpenGL rendering systems using Entity Component System (ECS) architecture and reusable components.
- Optimised rendering with sprite batching, shader and texture grouping, animation and GLSL shaders to reduce state changes and draw calls.
- Implemented camera controls, coordinate transformations, viewport resizing, and consistent 16:9 letterboxing for the game and editor.
- Integrated graphics tools into an ImGui editor and collaborated across engineering, design, and art disciplines to ship a playable game.

NYP Mobile Events Manager Application | Software Developer | 2019

- Built a full-stack events application for discovering, hosting and registering for events.
- Designed relational data models in MySQL and built Laravel/PHP REST APIs for authenticated access, data management, and application integration.
- Developed a React.js interface integrated with APIs to retrieve and update event information and registrations.

WORK EXPERIENCE

Sakura System Services | Systems Engineer | 2022 - 2023

- Developed and maintained client web applications based on business needs and technical requirements.
- Investigated application defects and troubleshot production issues to improve system reliability.
- Supported software deployment, system maintenance, and customer issue resolution.

Singapore Armed Forces | 3rd Sergeant | 2020 - 2022

- Led and supervised personnel, delegating responsibilities and coordinating team activities to achieve operational objectives.
- Demonstrated leadership, effective communication, and decision-making while managing training activities and personnel in high-pressure environments.

Sopra Steria | Android Developer Intern | 2019

- Developed Android applications in Android Studio using functional and client requirements.
- Collaborated on application troubleshooting, technical problem-solving, and improvements to development workflows.
- Prepared project documentation and technical presentations for internal stakeholders and client-facing discussions.
- Gathered user requirements and communicated project status with clients and engineering team members to support delivery.

EDUCATION

Singapore Institute of Technology & DigiPen Institute of Technology
Bachelor of Science with Honours in Computer Science in Real-Time Interactive Simulation
Expected Graduation: 2028

Nanyang Polytechnic
Diploma in Multimedia & Infocomm Technology
Graduated: 2020

CERTIFICATIONS

- **AI Singapore:** AI4I - Literacy in AI (Credential ID: 83957AC035-7F119FDAFC-156A3145B)

RELEVANT COURSEWORK

- **Machine Learning; Probability and Statistics; Algorithm Analysis**
- **Artificial Intelligence for Games; Data Structures; Linear Algebra**
- **Programming Massively Parallel Processors; Software Engineering Projects**